COMP 2711
Discrete Math Tools for Computer Science
Mock Midterm Examination 02

Practice material – not an official examination

Time allowed: 120 minutes Total: 100 points

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This closed-book practice exam consists of 10 problems. You may use one A4 single-sided handwritten cheat sheet.
- Answer all problems in the spaces provided. You may also use the backs of pages and the draft pages. Each problem starts on a new page; some answers require only a few lines.
- Give a proof, justification, or counterexample when requested. State any additional notation you introduce.
- Answers may be left in exact form using binomial coefficients, factorials, and $C(n,k)$ or $P(n,k)$. For example, $\binom{5}{3} + \binom{4}{2}$ is acceptable. Avoid nonstandard notation such as nCk or nPk.
- The separate solutions document contains derivations and marking notes. Attempt the exam before consulting it.

Problems	1	2	3	4	5	6	7	8	9	10	Total
Points	10	13	12	16	10	9	7	6	7	10	100
Score											

Problem 1: Propositional logic and proofs [10 points]

(a) **[3 points]** Construct a complete truth table for $(p \oplus q) \leftrightarrow \neg(p \leftrightarrow q)$. Classify it as a tautology, contradiction, or contingency.

(b) **[2 points]** For $p \rightarrow q$, write its converse, inverse and contrapositive. Which of these is always equivalent to the original?

(c) **[5 points]** Use contraposition to prove: for positive real numbers a, b, c, if $ab \leq c^2$, then $a \leq c$ or $b \leq c$. State the contrapositive explicitly.

Problem 2: Predicate logic [13 points]

(a) **[4 points]** The domain is all employees. $M(x, y)$ means y mentors x. Translate: (i) every employee has exactly one mentor; (ii) there is exactly one employee who mentors everyone. Each is worth 2 points.

(b) **[4 points]** Prove $(\forall x P(x)) \vee (\forall x Q(x)) \Rightarrow \forall x(P(x) \vee Q(x))$. Give a two-element countermodel to the reverse implication.

(c) **[5 points]** From $\forall x(P(x) \rightarrow Q(x))$ and $\forall x(Q(x) \rightarrow R(x))$, derive $\forall x(P(x) \rightarrow R(x))$. Name the inference rules and explain why generalization is allowed.

Problem 3: Sets and functions [12 points]

Use $\mathbb{N} = \{0, 1, 2, \ldots\}$.

(a) **[2 points]** Let $S = \{\varnothing, \{1\}\}$. List $\mathcal{P}(S)$ and find its cardinality.

(b) **[3 points]** Let $f, g : \mathbb{Z} \to \mathbb{Z}$ satisfy $f(x) = x+1$ and $g(x) = -x$. Compute $f \circ g$ and $g \circ f$, and determine whether the two compositions are equal.

(c) **[7 points]** Write T or F for each statement. No justification is required. Each item is worth 1 point.

(i) $\mathbb{N} \times \mathbb{N}$ is countably infinite.

(ii) The set of infinite binary sequences with exactly two ones is uncountable.

(iii) The set of binary strings of length at most 8 is finite.

(iv) $\mathbb{Z}$ and the positive even integers have equal cardinality.

(v) Every proper subset of an infinite set has smaller cardinality.

(vi) $[0, 1]$ is uncountable.

(vii) If $f : A \to B$ is a bijection, then $f^{-1} : B \to A$ is a function.

Problem 4: Short counting calculations [16 points]

Give exact answers. Factorials, powers, and binomial coefficients may be left unevaluated. No explanation is required unless requested.

(a) **[2 points]** A journey from A to C is either one of three direct roads or exactly one road from A to B followed by one from B to C. There are two A–B roads and four B–C roads. Count these journeys.

(b) **[2 points]** Count bit strings of length seven with exactly three ones and different first and last bits.

(c) **[2 points]** Count distinct strings formed by rearranging all letters of BALLOON, with equal letters indistinguishable.

(d) **[2 points]** Eight distinct files are assigned to labelled folders A, B, C, with respective sizes $2, 3, 3$. Within-folder order does not matter. Count the assignments.

(e) **[2 points]** Distribute nine indistinguishable tokens among four labelled boxes, with each box nonempty. Count the distributions.

(f) **[2 points]** Find the constant term in $(2x^2 - x^{-1})^9$.

(g) **[2 points]** Count right/up unit-step lattice paths from $(0, 0)$ to $(5, 4)$ that avoid $(2, 2)$.

(h) **[2 points]** From five chemists and four physicists, all distinct and in disjoint groups, select an unordered four-person committee containing exactly two chemists. Count the committees.

Problem 5: Combinatorial proof [10 points]

[10 points] For nonnegative integers m, n and $0 \leq r \leq m + n$, prove combinatorially that

$$\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}.$$

Use $\binom{a}{b} = 0$ when $b < 0$ or $b > a$.

Problem 6: Inclusion-exclusion [9 points]

[9 points] Six distinct jobs are assigned to three labelled servers A, B, C. Job 1 may use A or B only; the other jobs may use any server. Use inclusion–exclusion to count assignments in which every server is used.

Problem 7: Pigeonhole principle [7 points]

[7 points] Thirteen points lie in the closed rectangle $[0, 8] \times [0, 6]$. Prove that two are at distance at most $2\sqrt{2}$, and that four lie in some vertical strip of width two. The claims may use different points. Specify boundary conventions.

Problem 8: A nonuniform experiment [6 points]

The outcomes are $S = \{a, b, c, d\}$, with masses $k, 2k, 3k, 4k$ in that order. Define $X(a) = X(d) = 0$, $X(b) = 1$, $X(c) = 2$. Let $E = \{a, b\}$ and $F = \{a, c\}$.

(a) **[1 points]** Find k and verify that this is a probability mass function.

(b) **[1 points]** Find $P(E \cup F)$.

(c) **[1 points]** Find $P(X = 0)$.

(d) **[1 points]** Is X a deterministic function on S? Explain.

(e) **[2 points]** Are E, F independent? Show the required comparison.

Problem 9: Selection without replacement [7 points]

Choose urn A with probability $2/3$ and urn B with probability $1/3$. A contains two red and one blue labelled ball; B contains one red and two blue labelled balls. Draw two balls uniformly in order without replacement.

(a) **[3 points]** Find the probability that both draws are red.

(b) **[4 points]** Given exactly one red draw, find the probability that A was selected. Is this observation independent of the urn-selection event?

Problem 10: Random hat returns [10 points]

For $n \geq 2$, return n distinct hats to their owners using a uniformly random permutation. Let I_i indicate that owner i gets the correct hat, and let $H = \sum_{i=1}^{n} I_i$.

(a) **[2 points]** Find $P(I_i = 1)$ and $E(I_i)$.

(b) **[2 points]** Find $E(H)$. Does this step require independent indicators?

(c) **[2 points]** For $i \neq j$, find $P(I_i = I_j = 1)$ and test independence.

(d) **[4 points]** For arbitrary finite-space random variables U, V and real a, b, prove $E(aU + bV) = aE(U) + bE(V)$ directly from the definition.

Draft Paper

Mock examination working space.

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